San Francisco State University

Math 748

Theory & Applications of Statistical Machine Learning

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Final Project Report

High-Dimensional Gene Expression Classification

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1. Executive Summary

This project investigates the application of statistical machine learning techniques for classifying cancer types using the high-dimensional NCI60 gene expression dataset. The dataset presents a challenging scenario where the number of features (6,830 genes) greatly exceeds the number of samples (64 observations), resulting in a p >> n setting. Such conditions introduce fundamental statistical issues including overfitting, high variance in model estimates, instability in feature selection, and non-identifiability of classical models.

The analysis begins with exploratory data analysis, which reveals significant class imbalance across 14 cancer types. Several classes contain only one or two samples, making direct multi-class modeling unreliable. To address this, the study first focuses on a balanced binary classification problem using two well-represented cancer types (RENAL and NSCLC), each with 9 samples, ensuring a stable evaluation framework. Additional preprocessing steps, including feature standardization and variance analysis, confirm that the dataset contains meaningful variability and is suitable for modeling without feature filtering. Principal Component Analysis (PCA) further shows that while some low-dimensional structure exists (with the first 10 components explaining approximately 46% of the variance), complete separation between classes is not achieved.

Multiple modeling approaches are implemented and compared in the binary setting. A baseline Logistic Regression model achieves a mean cross-validation accuracy of 73.33%, but exhibits high variability across folds, indicating instability due to overfitting. Ridge Regression (L2 regularization) improves coefficient stability but does not enhance predictive performance, maintaining the same accuracy (~73.33%) across all regularization strengths. In contrast, Lasso Regression (L1 regularization) significantly improves performance, achieving a peak cross-validation accuracy of 85% after hyperparameter tuning. This improvement is driven by its ability to perform feature selection, reducing the feature space from 6,830 genes to only 54 informative features, thereby removing noise and improving generalization.

Dimensionality reduction using PCA combined with Logistic Regression is also evaluated. While PCA reduces dimensionality and stabilizes the model, its performance remains similar to baseline methods (~73%), as it transforms features rather than eliminating irrelevant ones. This highlights a key insight: feature selection is more effective than feature transformation in high-dimensional, low-sample-size settings.
To extend the analysis, the study explores multi-class classification using a filtered dataset consisting of 8 cancer types and 57 samples. In this more complex setting, baseline multi-class Logistic Regression achieves an accuracy of approximately 66.8%, while Lasso

Regression achieves a slightly lower accuracy of around 65.1% at optimal tuning. The decrease in performance compared to the binary case (85%) reflects the increased complexity of distinguishing multiple cancer types. Evaluation using confusion matrices and classification reports shows that model performance varies significantly across classes, with some cancer types (e.g., COLON, LEUKEMIA) being easily seperable, while others exhibit substantial overlap and are frequently misclassified.

Overall, the results demonstrate that feature selection is critical for high-dimensional data analysis. Lasso Regression provides the best balance between accuracy, interpretability, and generalization by identifying a sparse subset of informative features. However, the study also highlights the limitations of small sample sizes and the increased difficulty of multi-class classification in high-dimensional settings.

In conclusion, this project shows that in p >> n problems, models that effectively reduce dimensionality through feature selection outperform those that rely solely on coefficient shrinkage or feature transformation. The findings emphasize the importance of regularization, careful model evaluation, and problem simplification when working with high-dimensional biological data.

2. Introduction: Background and Problem Statement

Advancements in biotechnology and high-throughput sequencing technologies have led to the generation of large-scale gene expression datasets, enabling researchers to study complex biological processes at a molecular level. One important application of such data is the classification of cancer types based on gene expression profiles. Accurate classification can support early diagnosis, improve treatment planning, and contribute to a better understanding of disease mechanisms.

However, gene expression datasets introduce significant challenges for statistical learning methods. In many real-world biological datasets, including the NCI60 dataset used in the study, the number of features (genes) is extremely large compared to the number of available samples. Specifically, the dataset contains 6,830 gene expression measurements for only 64 samples, resulting in a high-dimensional setting where p >> n. This imbalance creates fundamental difficulties for traditional modeling approaches.

In such high-dimensional settings, classical statistical methods such as logistic regression become ill-posed. The design matrix lacks full rank, making parameter estimation unstable or non-unique. As a result, models may overfit the training data, capturing noise rather than meaningful patterns. This leads to poor generalization performance when applied to new data. Additionally, small samples sizes further amplify these issues, increasing the variance of model estimates and making results highly sensitive to small changes in the training data.

Another key challenge arises from the structure of the dataset itself. The NCI60 dataset includes 14 different cancer types, but the distribution of samples across these classes is highly imbalanced. While some cancer types, such as RENAL and NSCLC, have relatively more observations, several others contain only one or two samples. This imbalance makes multi-class classification difficult, as models may not have enough data to learn reliable patterns for underrepresented classes.

To address these challenges, this project adopts a structured approach to modeling. The analysis begins with a simplified binary classification problem using two well-represented cancer types (RENAL and NSCLC), allowing for more stable model training and evaluation. This controlled setup provides a foundation for understanding model behavior in high-dimensional settings. Once insights are established, the analysis is extended to a multi-class classification problem using a filtered subset of cancer types with sufficient sample sizes.

The primary objective of this project is to evaluate how different statistical learning techniques perform under high-dimensional conditions. In particular, the study focuses on understanding the role of regularization and dimensionality reduction in improving model stability and predictive performance. Methods such as Ridge Regression (L2 regularization), Lasso Regression (L1 regularization), and Principal Component Analysis (PCA) are explored and compared to determine their effectiveness in handling the challenges of high-dimensional data.

Ultimately, this project aims to answer a broader question: how can reliable and interpretable models be built when the number of predictors vastly exceeds the number of observations? By systematically analyzing different approaches and comparing their performance, this study provides insights into best practices for modeling high-dimensional biological data.

3. Research Questions

This project is guided by a set of research questions aimed at understanding the behavior of statistical learning methods in high-dimensional settings, particularly in the context of gene expression data. These questions are designed to evaluate both model performance and the underlying statistical challenges associated with p >> n problems.

The primary research questions are as follows:

1. How do classification models behave in a high-dimensional setting where the number of features greatly exceeds the number of samples (p >> n)?
   This question addresses the fundamental challenge of the dataset. In such settings, models are prone to overfitting and instability. Understanding this behavior is essential for selecting appropriate modeling techniques.
2. Does regularization improve model stability and generalization performance?
   Regularization methods such as Ridge (L2) and Lasso (L1) are commonly used to control model complexity. This question explores whether introducing penalties on model coefficients can reduce overfitting and lead to more reliable predictions.
3. How does feature selection impact predictive performance in high-dimensional data?
   Given that many features may be irrelevant or noisy, this question investigates whether selecting a subset of informative features (as done by Lasso) improves model accuracy and interpretability compared to using all features.
4. Can dimensionality reduction techniques such as Principal Component Analysis (PCA) effectively capture meaningful structure in the data?
   This question evaluates whether transforming the data into a lower-dimensional space helps improve model performance, or whether important information is lost in the process.
5. How does model performance differ between binary and multi-class classification settings?
   The project first analyzes a simplified binary classification problem and then extend to a multi-class setting. This question examines how increasing problem complexity affects model accuracy, stability, and interpretability.
6. What trade-offs exist between model complexity, interpretability, and predictive performance?
   This question focuses on the bias-variance tradeoff and aims to understand how different modeling approaches balance simplicity (fewer features), accuracy, and robustness.

4. Dataset Description

The dataset used in this project is the NCI60 gene expression dataset, a widely studied benchmark in bioinformatics for cancer classification tasks. It consists of gene expression measurements collected from multiple cancer cell lines, where each sample represents a specific cancer type.

The dataset contains:

- 64 samples (observations)
- 6,830 gene expression features (predictors)
- 14 distinct cancer types (classes)

The results in a high-dimensional setting ($p >> n$), where the number of features is more than 100 times the number of samples ($p/n \sim 106.7$). Such a structure makes the dataset particularly suitable for studying the behavior of statistical learning models under high-dimensional conditions.

Class Distribution and Modeling Implications

The distribution of samples across cancer types is highly imbalanced. A few classes, such as RENAL and NSCLC, contain the largest number of samples (9 each), while several other classes have only one or two observations.

This imbalance creates significant challenges for multi-class classification:

- Models may not learn meaningful patterns for underrepresented classes
- Cross-validation folds may lack representation from smaller classes
- Performance metrics may become unreliable

To address this, the project adopts a two-stage modeling strategy.

Binary Classification Dataset

In the first stage, the dataset is restricted to a binary classification problem using two well-represented classes:

- RENAL (9 samples)
- NSCLC (9 samples)

This results in:

- 18 samples
- 6,830 features
- Balanced class distribution (9 vs 9)

Multi-Class Dataset (Filtered)

In the second stage, the analysis is extended to a multi-class classification problem by selecting only cancer types with sufficient sample sizes (at least 5 observations).

The filtered dataset includes:

- 57 samples
- 6,830 features
- 8 cancer types:
  RENAL, NSCLC, MELONOMA, BREAST, COLON, OVERIAN, LEUKEMIA, CNS

Overall, the dataset provides a realistic and challenging test case for statistical learning due to :

- Extremely high dimensionality
- Limited sample size
- Class imbalance

These characteristics make it ideal for studying the effects of regularization, feature selection, and dimensionality reduction in classification tasks.

5. Statistical Challenges

The NCI60 gene expression dataset presents several fundamental statistical challenges that arise due to its high-dimensional structure and limited sample size. These challenges significantly impact model performance, stability, and interpretability, and motivate the use of specialized techniques such as regularization and dimensionality reduction.

5.1 High-Dimensionality (p >> n)

The dataset contains 6,830 features and only 64 samples, resulting in high-dimensional setting where the number of predictors far exceeds the number of observations.

In such settings:

- The design matrix X is not full rank
- The matrix $X^T X$ becomes singular
- Classical estimation methods (e.g., maximum likelihood in logistic regression) may not have a unique or stable solution

5.2 Risk of Overfitting

With thousands of features and very few samples, models have excessive flexibility. This allows them to memorize the training data rather than learn meaningful patterns.

In extreme cases:

- Training error can approach zero
- The model captures noise instead of signal
- Performance on unseen data deteriorates significantly
-

5.3 Small Sample Size

The dataset has only 64 observations, and the binary classification setup further reduces this to 18 samples.
This leads to:

- High variance in parameter estimates
- Sensitivity to small changes in training data
- Unstable cross-validation results

5.4 Class Imbalance

The dataset includes 14 cancer types, but the number of samples per class varies widely. Some classes have only one observation, while others have up to nine.
This creates challenges such as:

- Bias toward well-represented classes
- Poor learning for rare classes
- Unreliable evaluation metrics

5.5 Feature Redundancy and Noise

With thousands of gene expression features:

- Many features are likely irrelevant or redundant
- Some features may be highly correlated
- Multiple feature subsets may produce similar predictions

5.6 Instability of Feature Selection

In high-dimensional settings with small samples sizes:

- Feature selection methods (like Lasso) may produce different subsets of features across runs
- Multiple equally good solutions may exist

5.7 Multiple-Class Complexity

When extending from binary to multi-class classification:

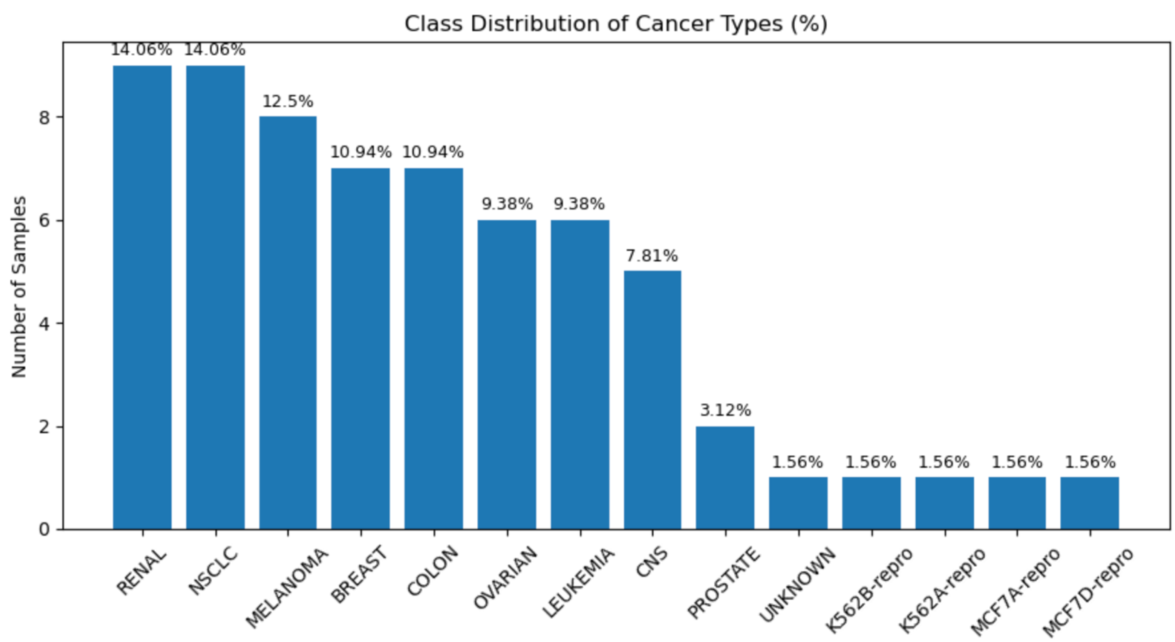
- The number of decision boundaries increases
- Classes may overlap in features space
- More features may be required to distinguish between classes

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the structure, variability, and underlying patterns in the NCI60 gene expression dataset. This step is essential in high-dimensional settings, as it informs preprocessing decisions and guides the choice of appropriate modeling techniques.

6.1 Class Distribution Analysis

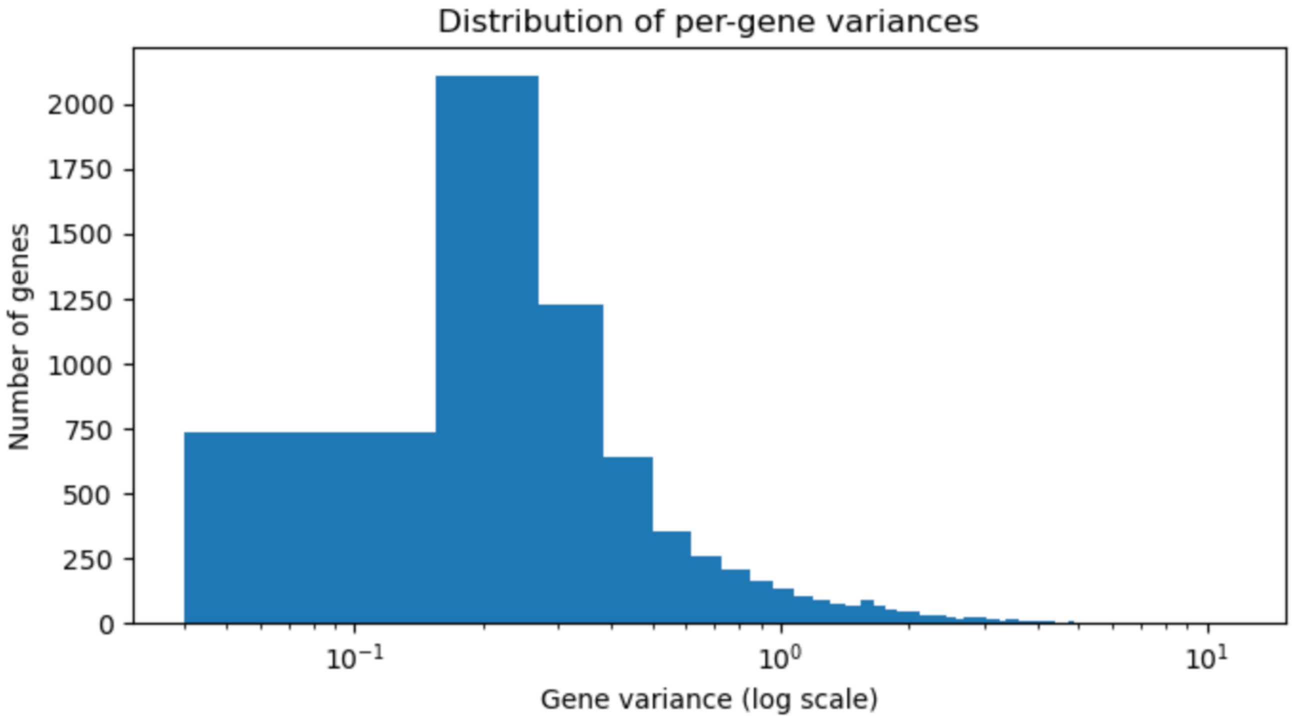
The dataset consists of 14 cancer types with a highly imbalanced distribution of samples. The largest classes, RENAL and NSCLC, each contain 9 samples, while several classes contain only one or two observations. This imbalance introduces significant challenges for multi-class classification, including biased learning, unreliable evaluation metrics, and instability in cross-validation.



Due to this imbalance, a two-stage modeling strategy was adopted. First, a binary classification problem was constructed using the two well-represented classes (RENAL and NSCLC) to ensure balanced and stable model evaluation. Subsequently, a filtered multi-class dataset was created by selecting only those cancer types with at least five samples, resulting in a more reliable multi-class classification setup.

6.2 Pre-Gene Variance Analysis

To evaluate the informativeness of features, the variance of each gene across sample was computed. The analysis showed that the minimum variance was approximately 0.0399, and no genes exhibited near-zero variance under multiple thresholds. The distribution of gene variances (on a logarithmic scale) revealed a broad spread, indicating that most genes contain meaningful variability.




```
Per-gene variance summary:
Number of genes: 6830
Min variance:      3.9927e-02
25% percentile:    2.0566e-01
Median variance:   3.1436e-01
Mean variance:     6.1279e-01
75% percentile:    6.3460e-01
Max variance:      1.1539e+01
```

This result suggests that the dataset does not contain redundant or constant features that need to be removed. Therefore, no variance-based feature filtering was applied, and all features were retained for further analysis.

6.3 Feature Standardization

All gene expression features were standardized to have a mean of zero and a standard deviation of one. This step is critical for ensuring that all features contribute equally to the model, particularly when applying regularization techniques such as Ridge and Lasso regression.

Standardization also ensures that the penalty terms in regularized models are applied uniformly across all features. Additionally, it is a necessary preprocessing step for Principal Component Analysis (PCA), which requires centered data to compute meaningful components.

```
Scaled data shape: (64, 6830)

Mean of first 5 features after scaling:
[-1.73472348e-17  1.47451495e-17 -5.09575021e-17 -4.85722573e-17
 -2.42861287e-17]

Std of first 5 features after scaling:
[1. 1. 1. 1. 1.]
```

6.4 Principal Component Analysis (PCA)

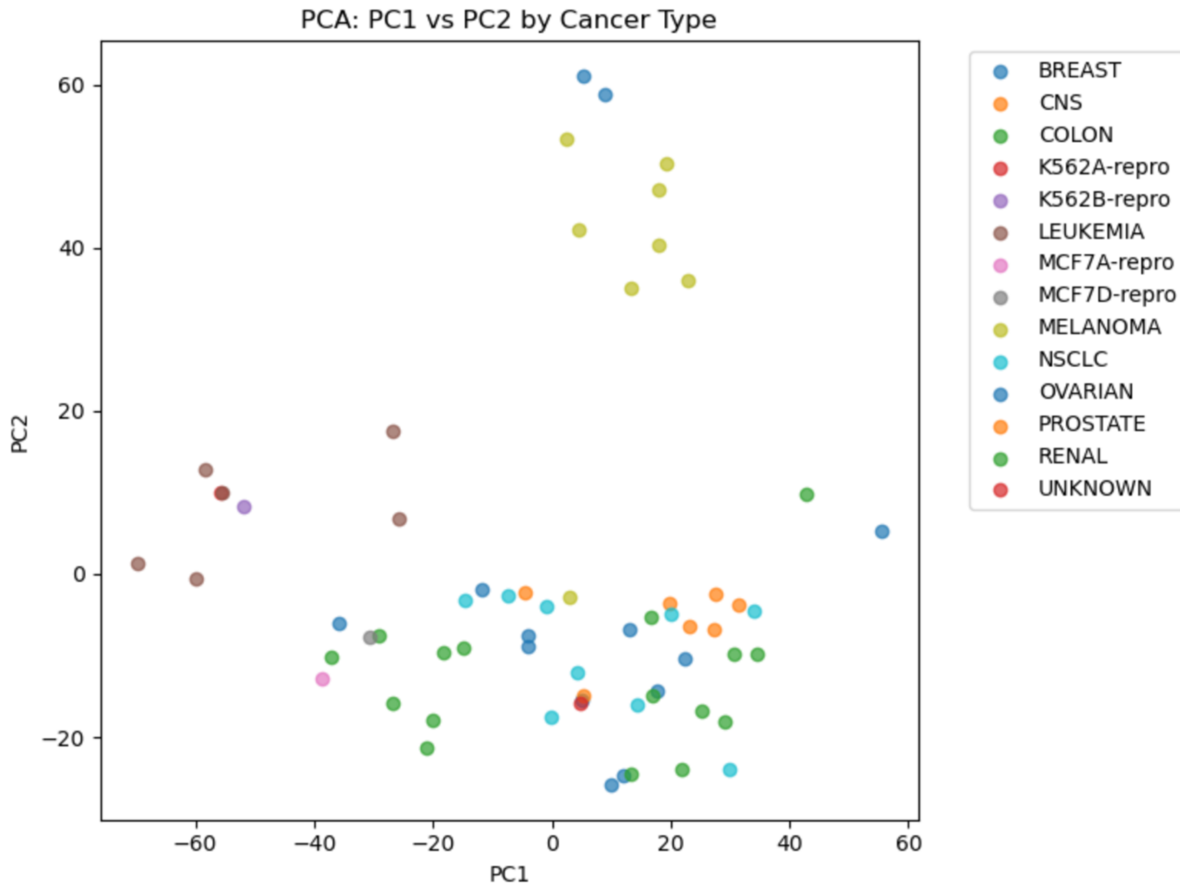
Principal Component Analysis was performed to explore the underlying structure of the dataset and assess the potential for dimensionality reduction. The results showed that the first principal component explains approximately 11.36% of the total variance, while the first ten components together explain about 46.13% of the variance.

```
Variance explained by first 10 components:
PC1: 0.1136
PC2: 0.0676
PC3: 0.0575
PC4: 0.0425
PC5: 0.0373
PC6: 0.0362
PC7: 0.0307
PC8: 0.0269
PC9: 0.0253
PC10: 0.0238

Cumulative variance explained by first 10 PCs: 0.4613
```

A scatter plot of the first two principal components revealed partial clustering of certain cancer types, such as MELANOMA and LEUKEMIA, indicating some degree of separability. However, significant overlap was observed among several other classes, suggesting that the dataset is not perfectly separable in low-dimensional space.

These findings indicate that while the dataset contains meaningful structure, dimensionality reduction alone may not be sufficient for accurate classification. Instead, additional techniques such as regularization and feature selection are required to effectively capture the underlying patterns.



The exploratory analysis highlights several important characteristics of the dataset:

- The dataset is highly imbalanced, requiring careful selection of classes for stable modeling
- Gene expression features exhibit meaningful variability, eliminating the need for variance-based filtering
- The data contains underlying structure but is not perfectly separable
- Dimensionality reduction captures partial structure but does not fully resolve classification challenges

Overall, the EDA results emphasize the need for regularization and feature selection techniques to handle the high-dimensional nature of the dataset and improve model performance.

7. Methodology

This section outlines the modeling approach adopted to address the high-dimensional classification problem. Given the challenges associated with p >> n settings, a structured methodology was designed to evaluate multiple statistical learning techniques and compare their effectiveness.

7.1 Overall Modeling Strategy

The analysis was conducted in two stages:

- **Stage 1: Binary Classification (RENAL vs NSCLC)**
  A simplified and balanced dataset was used to establish a stable modeling framework and understand model behavior under high-dimensional conditions.
- **Stage 2: Multi-Class Classification (Filtered Dataset)**
  The analysis was extended to a more realistic setting using multiple cancer types with sufficient sample sizes to evaluate model scalability and performance.

All models were evaluated using 5-fold cross-validation to ensure reliable performance estimate despite the small sample size.

7.2 Baseline Model: Logistic Regression

A standard Logistic Regression model without regularization was implemented as a baseline. This model serves as a reference point for evaluating the impact of regularization and dimensionality reduction techniques.

Key characteristics:

- No penalty term applied
- High flexibility in fitting data
- Evaluated using cross-validation

7.3 Ridge Regression (L2 Regularization)

Ridge Regression introduces an L2 penalty on the magnitude of coefficients, shrinking them toward zero.

Key characteristics:

- Penalizes large coefficients
- Reduces variance and improves stability
- Retains all features in the model

7.4 Lasso Regression (L1 Regularization)

Lasso Regression applies an L1 penalty, which not only shrinks coefficients but also forces some to become exactly zero, effectively performing feature selection.

Key characteristics:

- Produces sparse models
- Eliminates irrelevant features
- Control model complexity

Hyperparameter tuning was performed using GridSearchCV to identify the optimal value of the regularization parameter (C).

7.5 PCA + Logistic Regression

Principal Component Analysis (PCA) was used as a dimensionality reduction technique before applying Logistic Regression.

Key characteristics:

- Transforms original features into orthogonal components
- Reduces dimensionality
- Retains maximum variance

Different numbers of principal components were tested to evaluate their impact on model performance.

7.6 Hyperparameter Tuning

Hyperparameter tuning was performed using GridSearchCV for both Ridge and Lasso models. The regularization parameter C was varied across a range of values:

$$C \epsilon [0.001, 0.01, 0.1, 1, 10, 100]$$

- Smaller values of C correspond to stronger regularization.
- Larger values of C correspond to weaker regularization.

7.7 Multi-Class Modeling Approach

For the multi-class extension:

- A filtered dataset with 8 cancer types was used
- Logistic Regression with multinomial setting was applied as a baseline
- Lasso Regression with One-vs-Rest (OvR) strategy was used for regularized modeling

Hyperparameter tuning was again performed to identify the optimal regularization strength.

7.8 Evaluation Strategy

All models were evaluated using:

- 5-fold cross-validation (primary metric: accuracy)
- Confusion matrix (for multi-class analysis)
- Classification report (precision, recall, F1-score)

The methodology systematically evaluates:

- Baseline performance without regularization
- Impact of coefficient shrinkage (Ridge)
- Effectiveness of feature selection (Lasso)
- Role of dimensionality reduction (PCA)
- Model scalability from binary to multi-class settings

This structured approach ensures a comprehensive understanding of how different techniques perform under high-dimensional constraints.

8. Model Training and Results Interpretation (Binary Classification)

This section presents the training results and performance of different models on the binary classification problem (RENAL vs NSCLC). All models were evaluated using 5-fold cross-validation to ensure reliable performance estimates.

8.1 Logistic Regression (Baseline Model)

The baseline Logistic Regression model achieved a mean cross-validation accuracy of **0.7333**, with fold-wise accuracies ranging from **0.50 to 100**.

Table 1: Cross-validation accuracy scores for Logistic Regression

CV accuracy score	0.5	0.75	0.75	1	0.67
Mean CV score	0.7333				

The results indicate that the model is able to capture some meaningful patterns in the data. However, the large variation across folds suggests that model is unstable and highly sensitive to changes in the training data. This instability is a direct consequence of the high-dimensional setting, where the model has excessive flexibility and tends to overfit.

8.2 Ridge Regression (L2 Regularization)

Ridge Regression achieved a mean cross-validation accuracy of approximately **0.7333** after tuning, with performance remaining nearly constant across different values of the regularization parameter.

Table 2: Ridge Regression performance for different values of C

C value	Mean test score
0.001	0.7333
0.01	0.7333
0.1	0.7333
1	0.7333
10	0.7333
100	0.7333

These results suggest that while Ridge Regression helps stabilize the model by shrinking coefficients, it does not improve predictive performance. This is because Ridge retains all features, including irrelevant ones, allowing noise to remain in the model. As a result, reducing coefficient magnitude alone is not sufficient in high-dimensional settings.

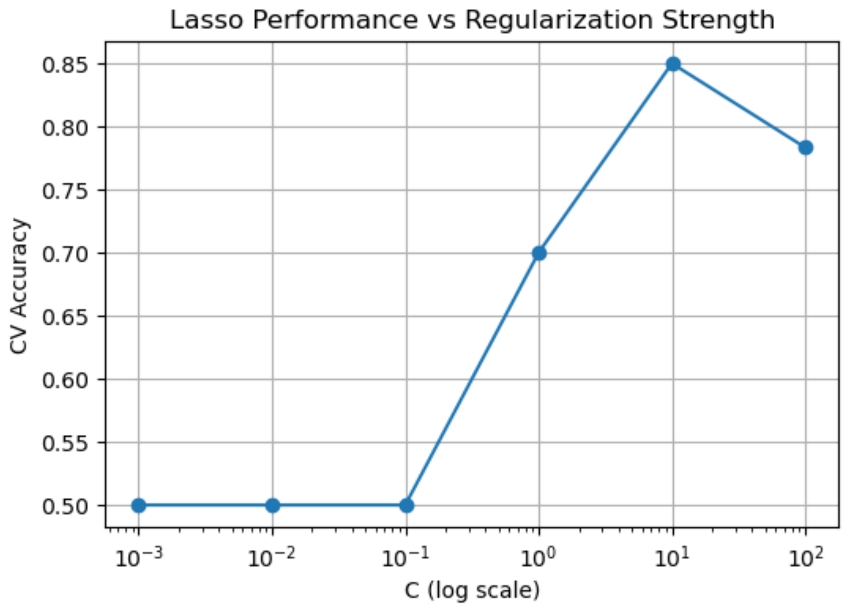
8.3 Lasso Regression (L1 Regularization)

Lasso Regression initially achieved a mean cross-validation accuracy of **0.70** with the default parameter setting. However, after hyperparameter tuning, the model reached a peak accuracy of **0.85** at an optimal value of C = 10.

Table 3: Lasso tuning results (C vs CV accuracy)

C value	CV accuracy
0.001	0.50
0.01	0.50
0.1	0.50
1	0.70
10	0.85
100	0.783

Figure: Lasso performance vs regularization strength (C)



The results show a clear pattern:

- Strong regularization (small C) leads to poor performance (~0.50), indicating underfitting
- Moderate regularization improves performance significantly
- Very weak regularization slightly reduces performance, indicating overfitting

This demonstrates the bias-variance tradeoff, where an optimal level of regularization balances model complexity and generalization performance.

The superior performance of Lasso indicates that feature selection plays a critical role in improving model accuracy in high-dimensional datasets.

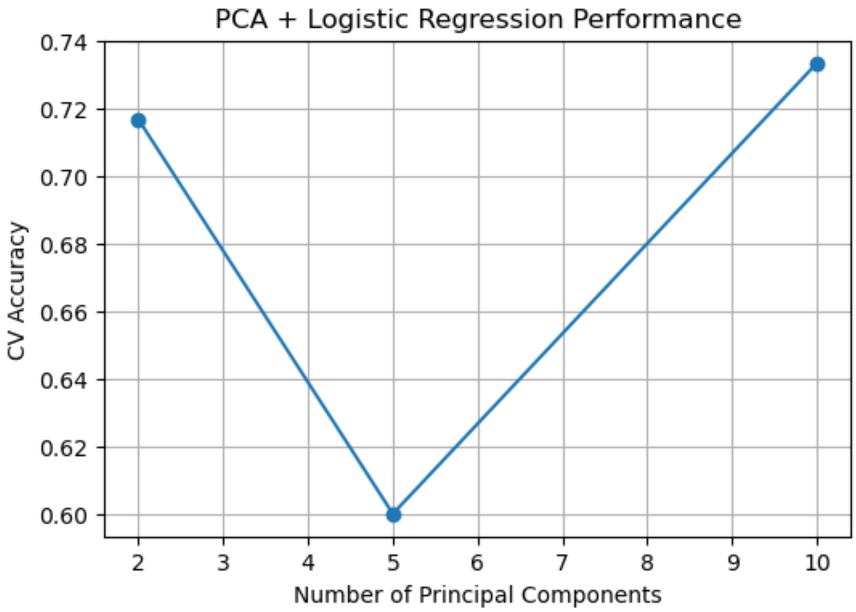
8.4 PCA + Logistic Regression

PCA combined with Logistic Regression achieved accuracies ranging from approximately **0.60 to 0.73**, depending on the number of principal components used.

Table 4: Number of PCA components vs CV accuracy

Number of components	CV accuracy
2	0.72
5	0.60
10	0.73
15	NaN

Figure: PCA components vs accuracy plot



The results show that:

- Using very few components captures limited information
- Increasing components improves performance up to a point
- Performance does not exceed baseline Logistic Regression

This suggests that while PCA reduces dimensionality, it does not effectively eliminate noise, as all original features contribute to the transformed components.

8.5 Key Observations

- Logistic Regression provides moderate accuracy but suffers from instability
- Ridge Regression improves stability but does not improve performance
- Lasso Regression achieves the highest accuracy by selecting relevant features
- PCA reduces dimensionality but does not outperform feature selection

Overall, the results demonstrate that feature selection is more effective than coefficient shrinkage or feature transformation in high-dimensional classification problems.

9. Feature Selection Analysis

Feature selection plays a critical role in high-dimensional datasets, where the number of features is extremely large compared to the number of samples. In this study, Lasso Regression was used to perform automatic feature selection by applying an L1 penalty, which forces many coefficients to become exactly zero.

9.1 Feature Selection using Lasso

Using the optimal regularization parameter C = 10, the final Lasso model selected 54 features (genes) with non-zero coefficients out of a total of 6,830 features.

Table 5: Feature Selection Summary

Total Features	6,830
Selected Features	54
Features Eliminated	6,776
Percentage Reduction	99%

This represents a reduction of more than **99%** of the original feature space, indicating that only a small subset of genes contributes meaningfully to the classification task.

9.2 Effect of Regularization on Feature Selection

The number of selected features varies significantly with the strength of regularization. As the parameter C changes, the model adjusts its complexity by including more or fewer features.

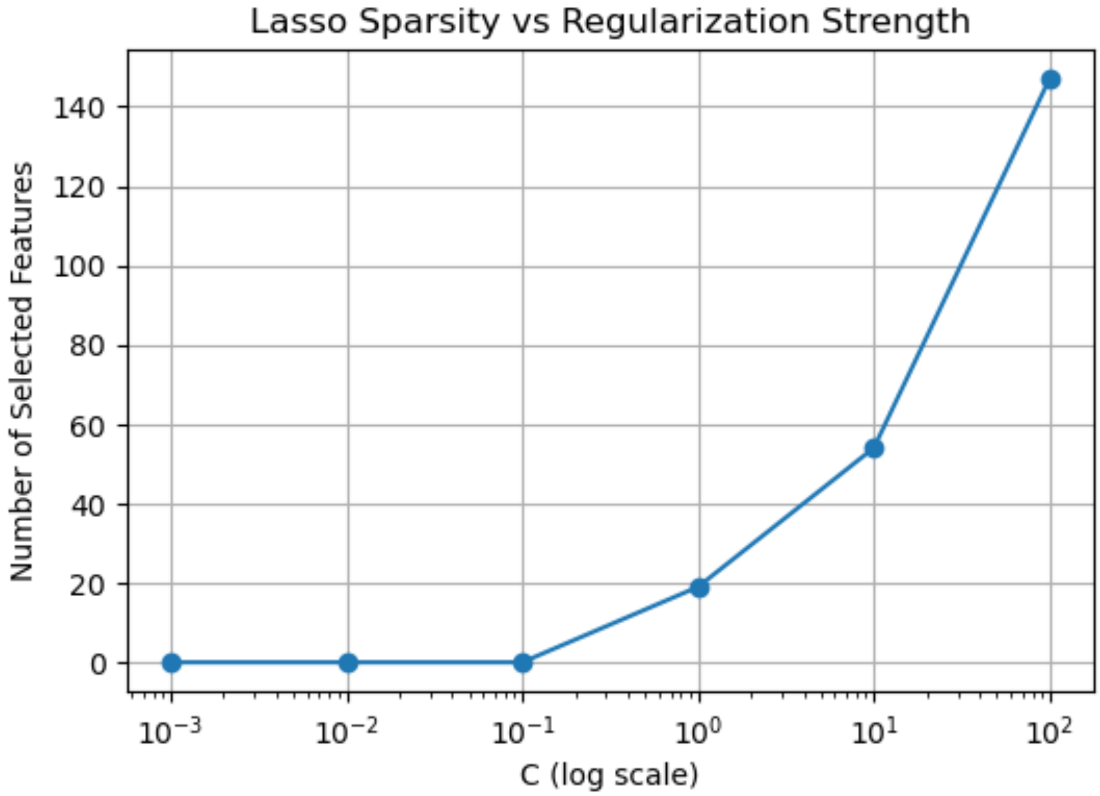
Observed pattern:

- For very small values of C, no features are selected, leading to complete underfitting
- As C increases, the model begins to select a small subset of features
- At C = 10, the model selects an optimal number of features (54)
- For very large values of C, the number of selected features increases significantly, indicating reduced regularization.

Table 6: Number of selected features for different values of C

C Value	Selected Features
0.001	0
0.01	0
0.1	0
1	19
10	54
100	147

Figure: Number of selected features vs C (log scale)



This behavior demonstrates how Lasso provides direct control over model complexity through feature selection.

9.3 Interpretation of Selected Features

The selected features represent genes that are most informative for distinguishing between the two cancer types. By eliminating irrelevant and redundant features, Lasso allows the model to focus on meaningful patterns in the data.

This improves:

- Model interpretability
- Generalization performance
- Resistance to overfitting

9.4 Stability of Feature Selection

An important observation is that the exact set of selected features may vary slightly across different runs. This is expected in high-dimensional settings due to:

- Small sample size
- High correlation among features
- Multiple feature subsets providing similar predictive performance

10. Model Evaluation

This section evaluates the performance and reliability of the models used in the binary classification task. While accuracy provides an overall measure of performance, additional analysis is required to assess model stability and generalization in high-dimensional settings.

10.1 Cross-Validation Performance

All models were evaluated using 5-fold cross-validation, which provides a more reliable estimate of model performance given the small sample size.

The results show that:

- Logistic Regression achieved moderate accuracy but showed high variability across folds
- Ridge Regression produced similar accuracy with slightly improved stability
- Lasso Regression achieved the highest accuracy after tuning
- PCA-based models showed moderate but consistent performance

10.2 Model Stability

A key aspect of model evaluation in high-dimensional settings is stability across different training splits.

The results indicate:

- Logistic Regression exhibits significant variability across folds, indicating instability
- Ridge Regression reduces variability slightly by shrinking coefficients
- Lasso Regression shows improved performance but still exhibits some variability due to feature selection
- PCA-based models are relatively stable but do not achieve high accuracy

10.3 Bias-Variance Tradeoff

The results across models illustrate the bias-variance tradeoff:

- Models with low regularization (e.g., Logistic Regression) have low bias but high variance, leading to overfitting
- Models with strong regularization (small C) have high bias and tend to underfit
- Models with moderate regularization (optimal C in Lasso) achieve the best balance

10.4 Generalization Performance

Generalization refers to how well a model performs on unseen data. In this study:

- Models that include all feature (Logistic, Ridge) tend to capture noise, limiting generalization
- Lasso improves generalization by removing irrelevant features
- PCA improves stability but does not eliminate noise, limiting its effectiveness.

11. Model Comparison & Discussion (Binary Classification)

This section compares the performance of all models used in the binary classification task and provides a detailed interpretation of their behavior in the high-dimensional setting.

11.1 Model Performance Comparison

The best cross-validation accuracies for each model are summarized below:

Table 7: Model comparison (Model vs Best CV Accuracy)

Model	Best CV Accuracy
Logistic Regression	0.7333
Ridge Regression (Tuned)	0.7333
Lasso Regression (Tuned)	0.8500
PCA + Logistic Regression	0.7333

11.2 Logistic Regression

The baseline Logistic Regression model achieved moderate accuracy but showed significant variability across folds. This indicates that while the model captures some signal, it is highly unstable due to overfitting.

Without regularization, the model is too flexible and fits noise in the data, leading to poor generalization.

11.3 Ridge Regression

Ridge Regression produced results similar to the baseline model, with no improvement in accuracy despite reducing coefficient magnitude.

Although Ridge stabilizes the model by shrinking coefficients, it does not eliminate irrelevant features. As a result, noise remains in the model, limiting its ability to improve performance.

11.4 Lasso Regression

Lasso Regression significantly outperformed all other models, achieving the highest accuracy of 0.85 after tuning.

The advantage of Lasso is its ability to perform feature selection. By removing irrelevant features, the model focuses only on informative variables, leading to better generalization and improved accuracy.

11.5 PCA + Logistic Regression

PCA combined with Logistic Regression achieved performance similar to the baseline model.

While PCA reduces dimensionality, it does not remove noise because all original features contribute to the principal components. This limits its effectiveness compared to Lasso.

The comparison clearly demonstrates that in high-dimensional settings:

- Models that include all features tend to overfit
- Methods that reduce dimensionality through feature selection perform better
- Simpler models with fewer but relevant features generalize more effectively

12. Extension to Multi-Class Classification

To extend the analysis beyond the binary classification setting, the study was expanded to a multi-class classification problem. This provides a more realistic evaluation of model performance, as real-world cancer classification involves multiple categories rather than just two.

12.1 Multi-Class Dataset Construction

Due to severe class imbalance in the original dataset, only cancer types with at least five samples were selected. This resulted in a filtered dataset with:

- 57 samples
- 6,830 features
- 8 cancer types: RENAL, NSCLC, MELANOMA, BREAST, COLON, OVARIAN, LEUKEMIA, CNS

This filtering ensures that each class has sufficient representation for stable training and evaluation.

12.2 Multi-Class Logistic Regression (Baseline)

A multinomial Logistic Regression model was applied as the baseline. The model achieved a mean cross-validation accuracy of approximately 0.6682, with noticeable variability across folds.

Table 8: Cross-validation scores for multi-class Logistic Regression

Multi-Class Logistic CV scores	0.5	0.75	0.7272	0.9090	0.4545
Mean CV accuracy	0.6682				

The model captures meaningful structure in the data, but performance is lower compared to the binary case. The increases number of classes introduced greater complexity, making classification more difficult.

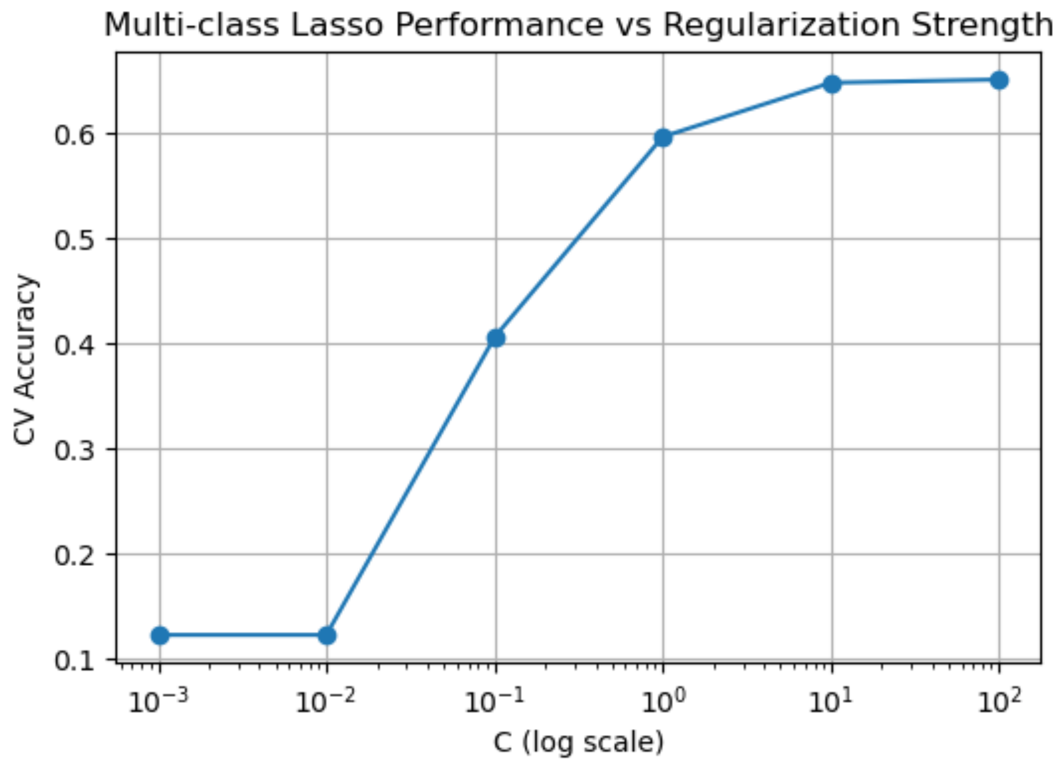
12.3 Multi-Class Lasso Regression

Lasso Regression was applied using a One-vs-Rest (OvR) strategy. After hyperparameter tuning, the model achieved a best cross-validation accuracy of approximately 0.6515 at C = 100.

Table 9: Multi-class Lasso tuning results (C vs CV accuracy)

C Value	**Mean CV Accuracy**
0.001	0.122727
0.01	0.122727
0.1	0.406061
1	0.596970
10	0.648485
100	0.651515

Figure: Multi-class Lasso performance vs C



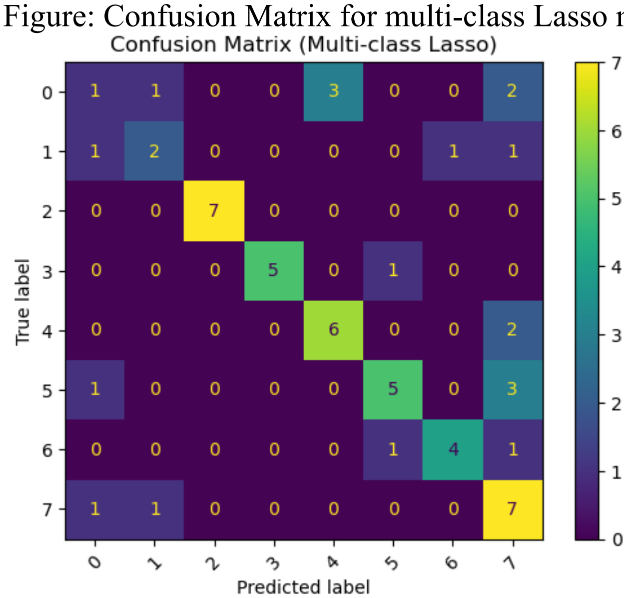
- Strong regularization leads to severe underfitting (very low accuracy)
- Increasing C improves performance as more features are included
- The optimal model requires weaker regularization compared to the binary case

This indicates that multi-class classification requires more features to capture complex class boundaries.

12.4 Confusion Matrix Analysis

The confusion matrix provides a detailed view of classification performance across different cancer types.

Figure: Confusion Matrix for multi-class Lasso model



The results show:

- Strong diagonal entries for certain classes (e.g., COLON, LEUKEMIA), indicating high classification accuracy
- Significant off-diagonal entries for others, indicating misclassification

Some cancer types are easily separable, while others have overlapping gene expression patterns, making them difficult to distinguish.

12.5 Classification Report

The classification report provides precision, recall, and F1-score for each class.

Table 10: Classification report for multi-class model

Type	Precision	Recall	F1-score	Support
BREAST	0.25	0.14	0.18	7
CNS	0.50	0.40	0.44	5
COLON	1.00	1.00	1.00	7
LEUKEMIA	1.00	0.83	0.91	6
MELANOMA	0.67	0.75	0.71	8
NSCLC	0.71	0.56	0.62	9
OVARIAN	0.80	0.67	0.73	6
RENAL	0.44	0.78	0.56	9
accuracy			0.65	57
macro avg	0.67	0.64	0.64	57
weighted avg	0.66	0.65	0.64	57

Key observations:

- Some classes achieve very high performance (e.g., COLON, LEUKEMIA)
- Other classes (e.g., BREAST, CNS) show lower performance
- Overall accuracy is approximately 65%, with balanced macro and weighted averages

Performance varies significantly across classes, indicating differences in class separability.

12.6 Binary vs Multi-Class Comparison

The transition from binary to multi-class classification results in:

- Accuracy decreasing from ~85% (binary) to ~65% (multi-class)
- Increased model complexity
- Greater overlap between classes
- Reduced stability and consistency

Multi-class classification is inherently more difficult due to the need to distinguish between multiple categories simultaneously.

Overall, the multi-class extension demonstrates that while the models capture meaningful structure in the data, classification becomes more challenging as the number of classes increases.

13. Discussion

This project provides a comprehensive analysis of statistical learning methods applied to a high-dimensional gene expression dataset. The results highlight several important insights regarding model behavior, feature selection, and the challenges associated with small sample sizes.

13.1 Impact of High-Dimensionality

The high-dimensional nature of the dataset ($p >> n$) plays a central role in model performance. Traditional models such as Logistic Regression struggle in this setting due to overfitting and instability. The results clearly show that models without proper constraints tend to capture noise rather than meaningful patterns.

High-dimensional datasets require specialized techniques to control model complexity and ensure reliable predictions.

13.2 Role of Regularization

Regularization was essential in improving model stability. Ridge Regression reduced variance by shrinking coefficients, but it did not improve accuracy because it retained all features. In contrast, Lasso Regression effectively addressed both variance and feature redundancy by performing feature selection.

Regularization alone is not sufficient; the ability to eliminate irrelevant features is crucial for improving performance.

13.3 Importance of Feature Selection

One of the most significant findings of this study is the importance of feature selection. Lasso reduced the feature space from 6,830 to just 54 features while achieving the highest accuracy.

This demonstrates that only a small subset of genes contains meaningful information for classification. Removing irrelevant features improves both model interpretability and generalization.

13.4 Feature Selection Stability

Although Lasso identifies important features, the exact set of selected genes may vary across different runs. This is due to the small sample size and high correlation among features.

Individual features should be interpreted with caution, and conclusions should focus on overall model performance rather than specific selected variables.

13.5 Binary vs Multi-Class Performance

A clear difference was observed between binary and multi-class classification:

- Binary classification achieved high accuracy (~85%)
- Multi-class classification achieved lower accuracy (~65%)

The increase in the number of classes introduces greater complexity, requiring the model to distinguish between multiple overlapping patterns.

13.6 Model Comparison Insights

The comparison of different models reveals that:

- Logistic Regression suffers from overfitting
- Ridge improves stability but not performance
- PCA reduces dimensionality but retains noise
- Lasso provides the best performance through feature selection

Feature selection is more effective than coefficient shrinkage or feature transformation in high-dimensional settings.

13.7 Limitations of the Study

Despite strong results, several limitations exist:

- Extremely small sample size reduces reliability of estimates
- High-dimensional setting increases variance and instability
- Feature selection results may not be fully consistent
- Multi-class classification remains challenging due to class overlap

Overall, the project demonstrates that Lasso Regression provides the most effective approach by balancing accuracy, interpretability, and generalization in high-dimensional classification tasks.

14. Conclusion

This project investigated the application of statistical machine learning techniques for classifying cancer types using high-dimensional gene expression data from the NCI60 dataset. The primary challenge of this problem arises from the p >> n setting, where the number of features far exceeds the number of samples, leading to overfitting, instability, and difficulty in model estimation.

The analysis began with a binary classification problem (RENAL vs NSCLC) to establish a stable modeling framework. In this setting, Logistic Regression achieved moderate accuracy but exhibited high variability, indicating overfitting. Ridge Regression improved model stability but did not enhance predictive performance, as it retained all features and failed to eliminate noise.

Lasso Regression emerged as the most effective method, achieving the highest cross-validation accuracy of approximately 85%. This improvement was driven by its ability to perform feature selection, reducing the feature space from 6,830 features to only 54 informative genes. This demonstrates that removing irrelevant features is critical for improving model performance in high-dimensional datasets.

Dimensionality reduction using PCA was also explored. While PCA helped reduce dimensionality and stabilize the model, it did not outperform Lasso, as it transforms features rather than selecting the most relevant ones. This highlights the limitation of feature transformation in the presence of noisy and irrelevant variables.

The analysis was then extended to a multi-class classification problem involving 8 cancer types. In this more complex setting, model performance decreased to approximately 65%, reflecting the increased difficulty of distinguishing between multiple classes. Evaluation using confusion matrices and classification reports revealed that some cancer types are easily separable, while others exhibit significant overlap in gene expression patterns.

In conclusion, this project demonstrates that in high-dimensional, low-sample-size settings, models that effectively reduce the feature space through feature selection outperform those that rely solely on coefficient shrinkage or dimensionality reduction. Lasso Regression, in particular, proves to be a powerful and practical approach for such problems.

15. Future Work

While this study provides valuable insights into high-dimensional classification, several directions can be explored to further improve the analysis and extend the findings.

15.1 Extension to Full Multi-Class Classification

The current multi-class analysis was limited to cancer types with at least five samples. Future work can extend this to include all 14 cancer types, potentially using advanced techniques to handle class imbalance.

Potential improvement can be using methods such as class weighting or resampling to address imbalance and improve model performance.

15.2 Increasing Dataset Size

The small sample size is a major limitation in this study. Increasing the number of observations would:

- Reduce variance in model estimates
- Improve stability of feature selection
- Provide more reliable evaluation metrics

Improvement that can be made is incorporate additional dataset or combine multiple gene expression datasets.

15.3 Feature Stability Analysis

Lasso feature selection may vary across different training splits. Future work can focus on evaluating the consistency of selected features. So using repeated cross-validation or bootstrapping to identify stable feature across multiple runs would be greater improvement.

15.4 Elastic Net Regularization

Elastic Net combines both L1 (Lasso) and L2 (Ridge) penalties and can be more effective when features are highly correlated. Improvement we can make is evaluating elastic net to balance feature selection and coefficient shrinkage.

15.5 Advanced Machine Learning Models

More complex models can be explored to improve classification performance. These models may capture non-linear relationships that linear models cannot.

15.6 Biological Interpretation of Selected Features

The selected gene indices can be mapped to actual biological gene names to gain domain-specific insights. As improvement we can identify biological pathways associated with selected genes and analyze their relevance to cancer classification.